Compulsory Exercise

Topics in Life Insurance (Liv2), Block 3, 2026

March 14, 2026

Contents

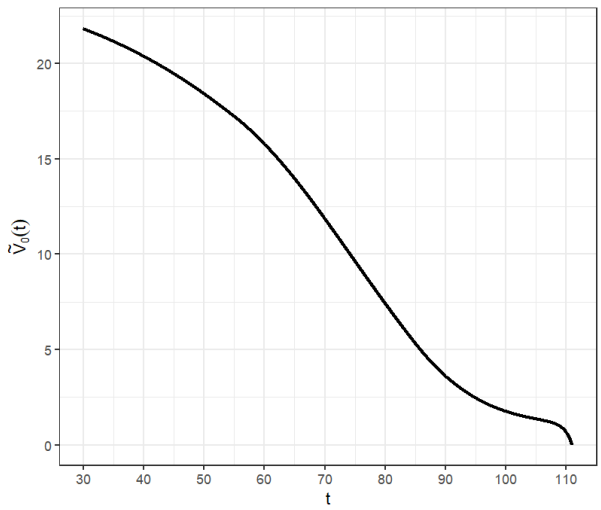
(i)

Using Lebniz' rule we obtain af ordinary backward differential equation for the reserve $\tilde{V}_0$. We have

$$\begin{aligned}
\frac{d}{dt}\tilde{V}_0(t) &= \int_t^\infty \frac{\partial}{\partial t} e^{-\int_t^s (\tilde{r}(u)+\tilde{\mu}(u))\,du}\,ds - 1 \\
&= \int_t^\infty e^{-\int_t^s (\tilde{r}(w)+\tilde{\mu}(w))\,dw}(\tilde{r}(t)+\tilde{\mu}(t))\,ds - 1 \\
&= (\tilde{r}(t)+\tilde{\mu}(t)) \int_t^\infty e^{-\int_t^s (\tilde{r}(w)+\tilde{\mu}(w))\,dw}\,ds - 1 \\
&= (\tilde{r}(t)+\tilde{\mu}(t))\tilde{V}_0(t) - 1,
\end{aligned}$$

with $\tilde{V}_0(111) = 0$.

Now, using the fourth order Runge-Kutta method we optain a numerical solution of $\tilde{V}_0(t)$. The result is shown via the plot below



Exercise (ii)

We simulate a single path of the unit-link account X from age 30 to 111 using the Euler scheme with step length $h = 1/10\,000$.

During the accumulation phase $t \in [30, m)$, the SDE is

$$\mathrm{d}X(t) = \Big[(\pi(t)\alpha + (1-\pi(t))r)X(t) + \theta L(t) - \mu(t)(KL(t) - X(t))\Big]\mathrm{d}t + \sigma\pi(t)X(t)\,\mathrm{d}W(t), \tag{1}$$

and during the decumulation phase $t \in [m, 111]$,

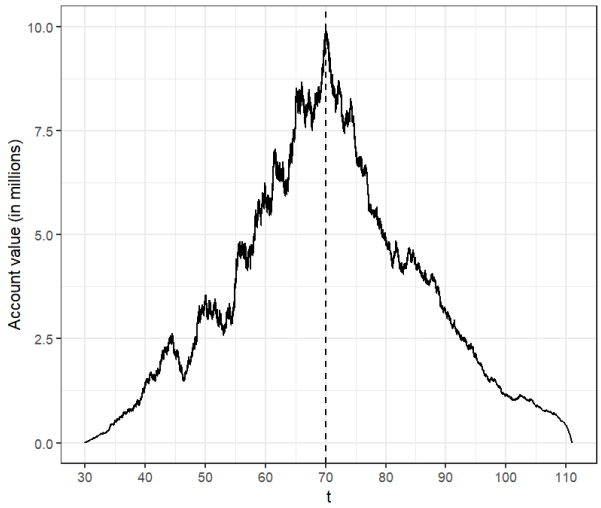
$$\mathrm{d}X(t) = \Big[(\pi(t)\alpha + (1-\pi(t))r)X(t) - \frac{X(t)}{\widetilde{V}_0(t)} + \mu(t)X(t)\Big]\mathrm{d}t + \sigma\pi(t)X(t)\,\mathrm{d}W(t). \tag{2}$$

The Euler update at each step $t_i \to t_{i+1} = t_i + h$ is

$$X(t_{i+1}) = X(t_i) + f(t_i, X(t_i))\,h + g(t_i, X(t_i))\sqrt{h}\,Z_i, \quad Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

(iii)

A simulation of the account is provided below.



We observe from our simulation that it begins from age $t = 30$ at the value of zero as demanded. Thereafter, the account accumulates until retirement $m = 70$ with a value of around 9.8 million (monetary units). Then the account de-cumulates until our assumed lifetime cut-off at $t = 111$. We also observe random fluctuates due to the fact that we model the risky asset via a Brownian motion.

Exercise (iv)

We argue that the market reserve equals the account value, i.e. $V^0(t, x) = x$.

The market reserve $V^0(t, x)$ is defined as the expected discounted future net cash flows from time t given that $X(t) = x$. Because the product is a unit-link contract, all liabilities are tied directly to the account value: the pension payout rate equals $X(t)/\widetilde{V}_0(t)$ and the death benefit is $-X(t)$ (i.e. the insurer simply pays out the account and retains no additional liability).

Formally, note that the SDE for X after retirement contains the term $\mu(t)(-X(t))\,\mathrm{d}t$, which exactly cancels the expected insurance payout upon death. Combined with the boundary condition $X(111) \approx 0$ (observed in the simulation, since the entire account is depleted over the lifetime), we have that the insurer's net liability at any time t is exactly $x = X(t)$.

Hence $V^0(t, x) = x$.

Exercise (v)

Let $Y(t) = b^0(t, X(t)) = X(t)/\widetilde{V}_0(t)$ for $t \geq m$. We show that Y is a generalised Geometric Brownian motion.

Applying Itô's formula to $Y(t) = X(t)/\widetilde{V}_0(t)$:

$$\mathrm{d}Y(t) = \frac{1}{\widetilde{V}_0(t)}\mathrm{d}X(t) + X(t)\,\mathrm{d}\left(\frac{1}{\widetilde{V}_0(t)}\right).$$

Using the decumulation SDE (2) for $\mathrm{d}X(t)$ and the relation $\frac{\mathrm{d}}{\mathrm{d}t}\widetilde{V}_0(t) = (\widetilde{r}(t)+\widetilde{\mu}(t))\widetilde{V}_0(t)-1$, after simplification one obtains

$$\mathrm{d}Y(t) = \Big(\pi(t)\alpha + (1-\pi(t))r - \widetilde{r}(t) + \mu(t) - \widetilde{\mu}(t)\Big)Y(t)\,\mathrm{d}t + \sigma\pi(t)Y(t)\,\mathrm{d}W(t), \tag{3}$$

with initial condition $Y(m) = X(m)/\widetilde{V}(m)$.

This is a linear SDE in Y, confirming that Y is a (generalised) Geometric Brownian motion.

Interpretation of the calculation basis. If the calculation basis satisfies $\widetilde{r}(t) = \pi(t)\alpha + (1-\pi(t))r$ and $\widetilde{\mu}(t) = \mu(t)$ (as in Table 2), then the drift of Y is zero and Y is a martingale. This means the expected annuity benefit is constant over time, i.e. the insured receives a flat expected payout profile. In general, the basis $(\widetilde{r}, \widetilde{\mu})$ determines how the payout profile is shaped: a higher $\widetilde{r}$ front-loads payments, while a lower $\widetilde{r}$ back-loads them.

Exercise (vi)

Using the GBM dynamics from Exercise (v), we simulate $N = 1000$ paths of the life annuity benefit process $Y(t)$ on $[m, 111]$ with Euler step $h = 1/100$.

The Euler update is

$$Y(t_{i+1}) = Y(t_i) + \mu_Y(t_i)\, Y(t_i)\, h + \sigma\pi(t_i)\, Y(t_i)\sqrt{h}\, Z_i, \quad Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1),$$

where $\mu_Y(t) = \pi(t)\alpha + (1 - \pi(t))r - \widetilde{r}(t) + \mu(t) - \widetilde{\mu}(t)$.

The common starting value is $Y(m) = X(m)/\widetilde{V}_0(m)$, where $X(m)$ is taken from the single path simulated in Exercise (ii).

Exercise (vii)

Since the calculation basis in Table 2 sets $\widetilde{r}(t) = \pi(t)\alpha + (1 - \pi(t))r$ and $\widetilde{\mu}(t) = \mu(t)$, the drift of Y vanishes and the expected benefit $\mathbb{E}[Y(t) \mid X(m)]$ is constant over the entire retirement period $[m, 111]$.

This is by design: the life-cycle unit-link product is constructed so that the insured receives a *level* expected payout throughout retirement. The actual realised benefits fluctuate around this level due to investment risk (the $\sigma\pi(t)$ diffusion term), but on average the benefit remains flat. This flat profile is a direct consequence of the calculation basis being chosen to match the true (market) discount rate and mortality.